Progress-check: Data Science Lab

Default of Credit Card Clients Prediction Model

Group 2

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GutHub Link for Progress Check

SCAN ME

DIRECT LINK

Dataset Introduction

Kaggle Dataset

Contains information on default payments, demographic factors, credit data, history of payment, and bill statements of credit card clients

Time & Place

Taiwan from April 2005 to September 2005

Variables

ID: ID of each client
LIMIT_BAL: Amount of given credit in NT dollars
SEX: Gender
EDUCATION
MARRIAGE: Marital status
AGE: Age in years
PAY_*X*: Repayment status in ***X***/2005
BILL_AMT*X*: Amount of bill statement in ***X***/2005
PAY_AMT*X*: Amount of previous payment in ***X***/2005
DEFAULT: Credit Default dummy

Research Questions: Foundation

1

Do such variables as sex, age, education and marital status help predict the probability of default? ***(Demographics Path)***

- age can mean more experience and more savings -> less probability of default - marriage can help mitigate the default due to common savings
- education can mean more financial awareness -> less probability of default

2

Do previous loan payments help predict probability of default? ***(Dynamic Path)***

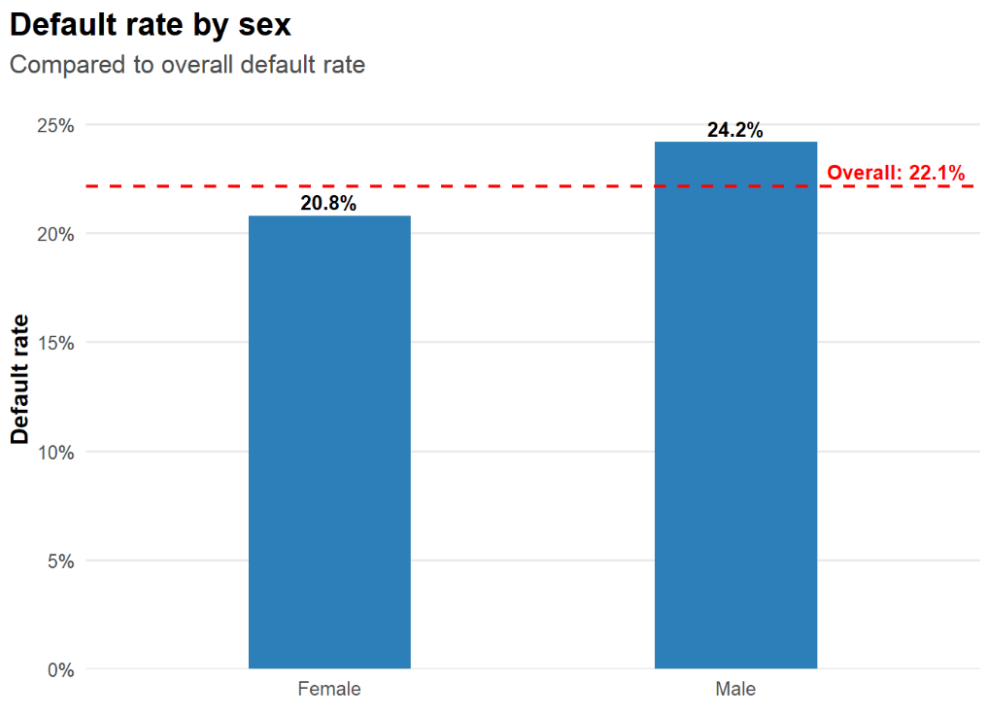
- diligent payments -> signal of low prob-ty of default
- or do people pay whenever it is convenient, and past payment are not good indicators?

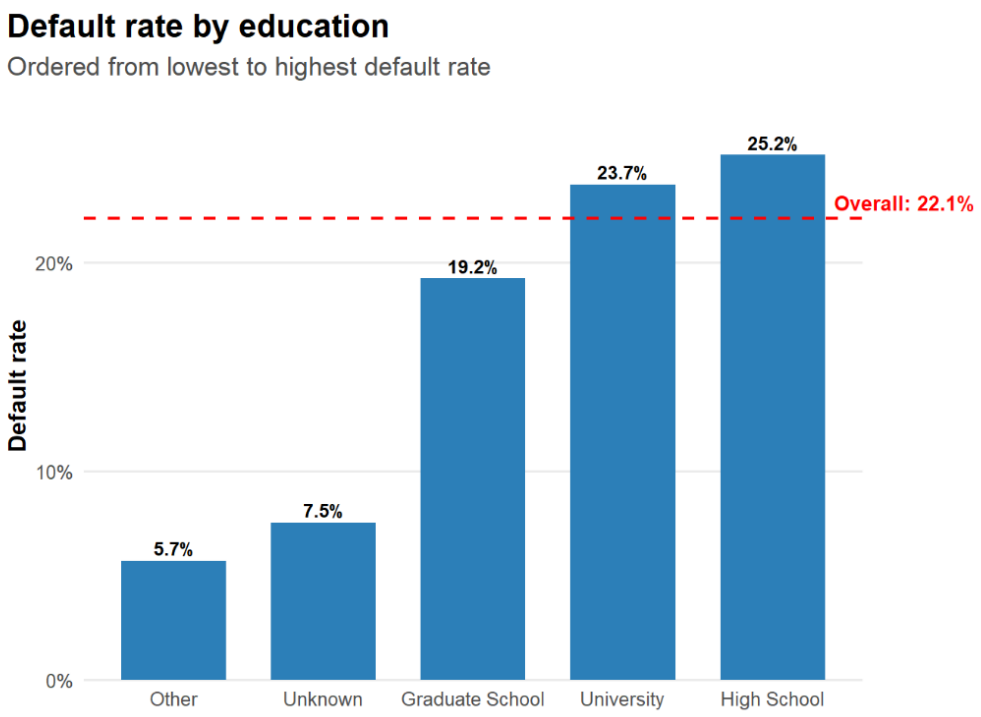
3

Does the loan amount serve as a default indicator? ***(Volume Path)***

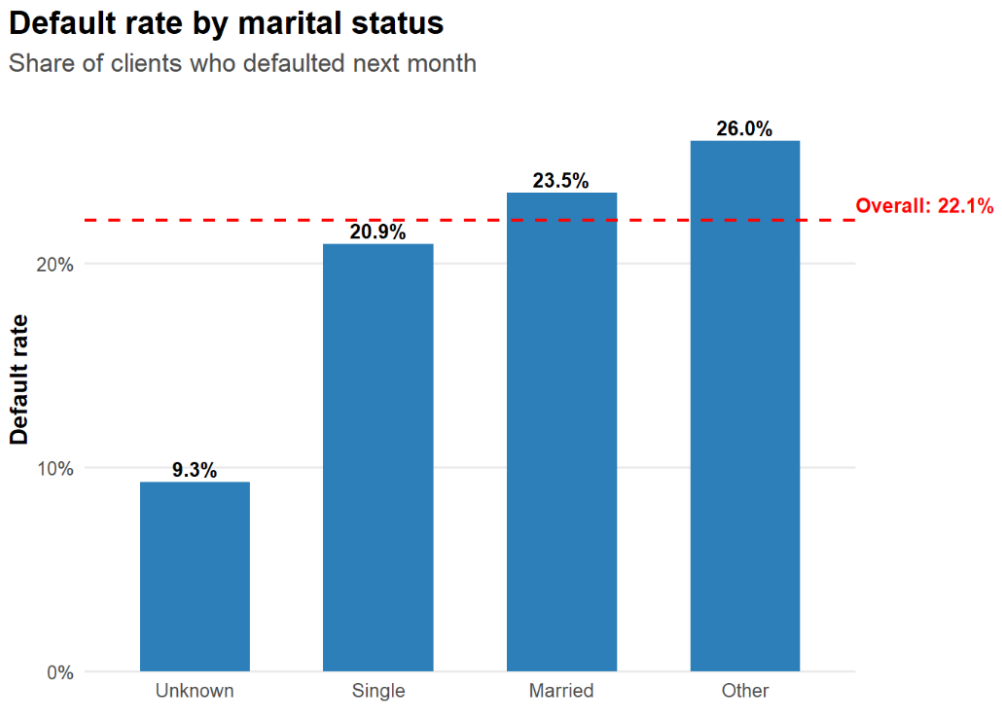
- more debt -> higher default rate?
- loan amount categories?

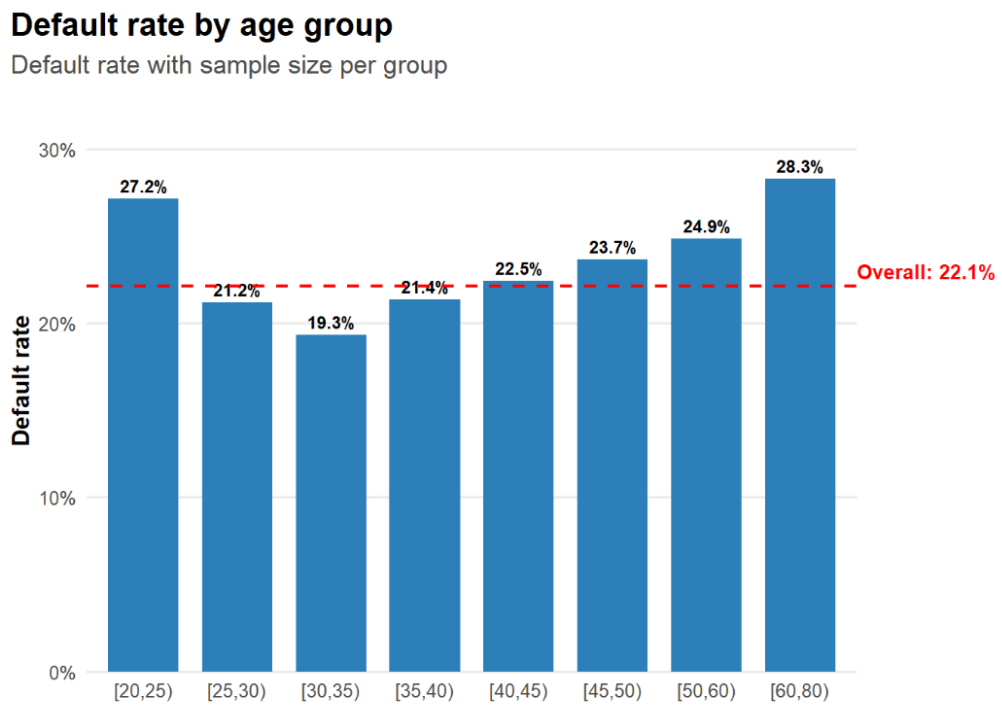
Dataset Visualization: Demographics and Default Rate



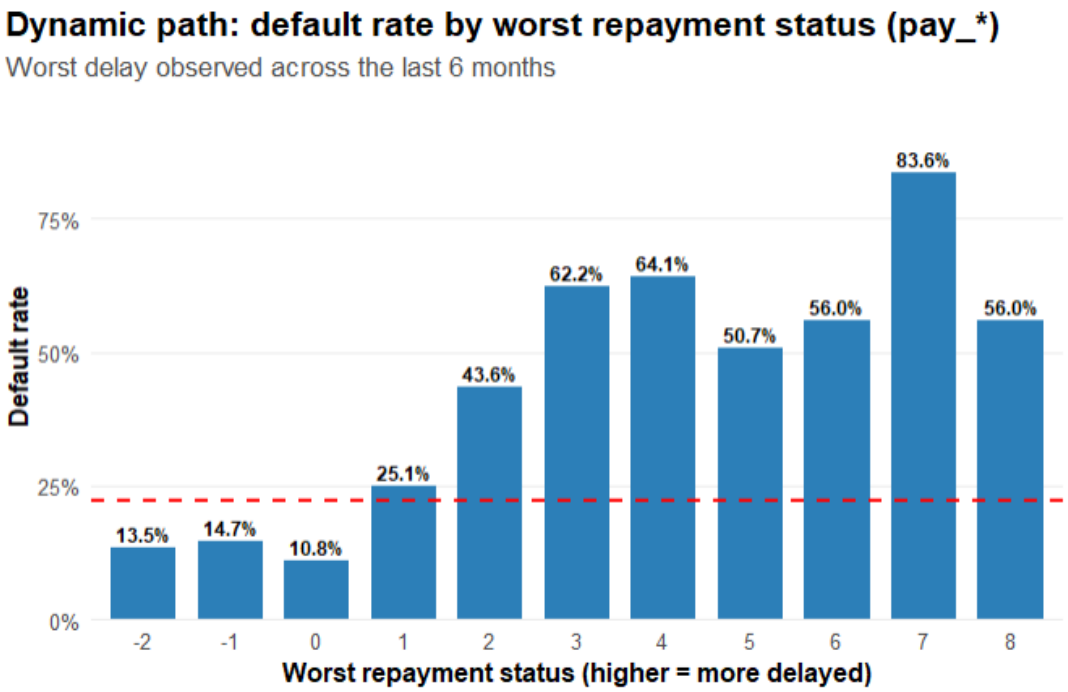


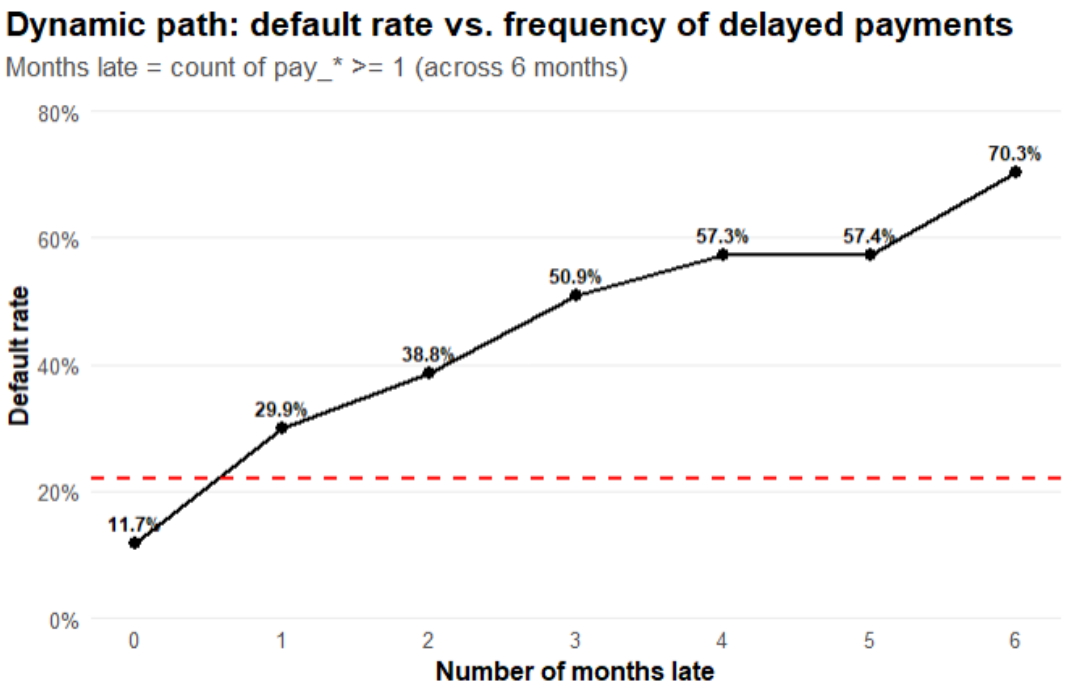
Dataset Visualization: Demographics and Default Rate



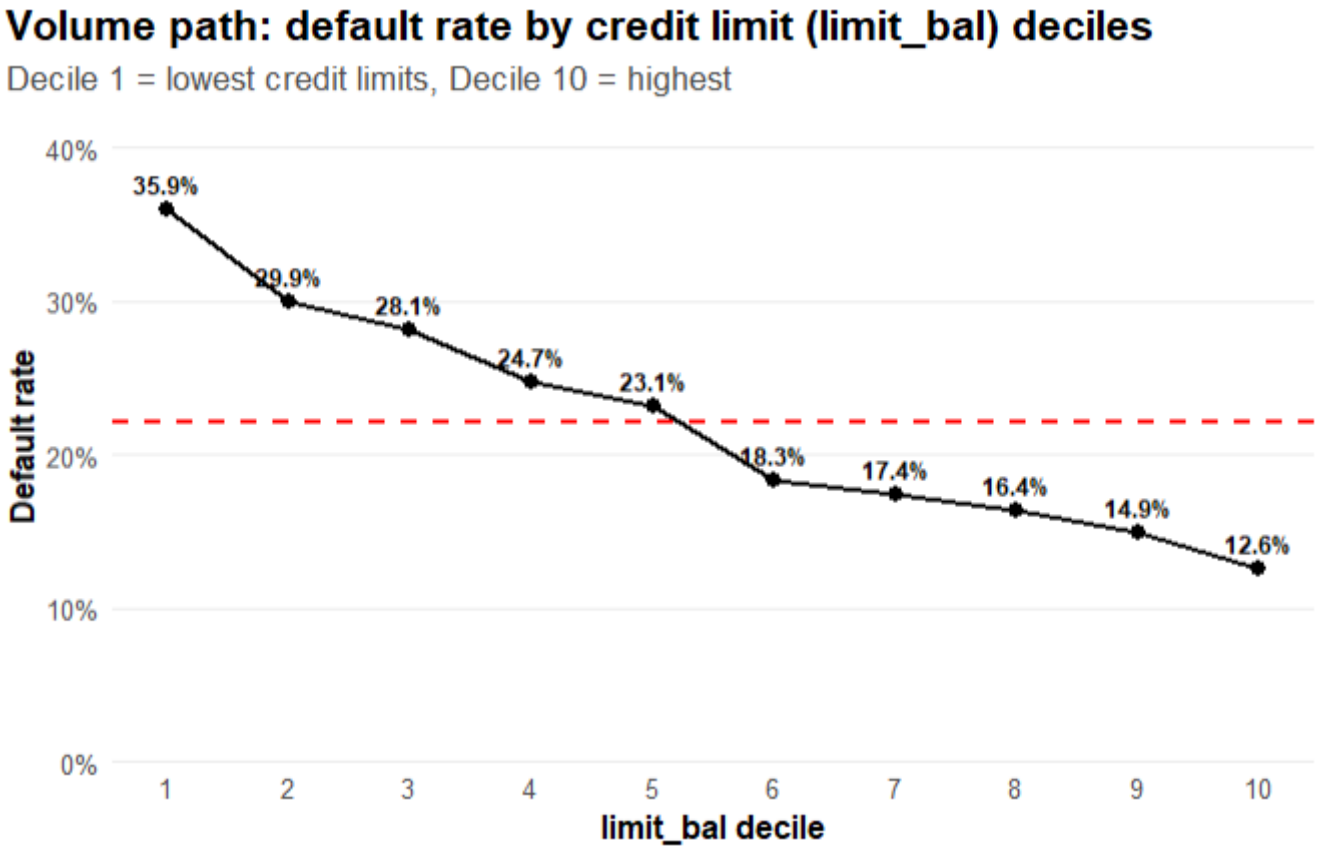


Dataset Visualization: Dynamic Path





Dataset Visualization: Volume Path



RQ Finalization: variable correlation check

	default_num	limit_bal	avg_repay_status	age	sex_male	education_level	married
married	0.03	0.1	-0.03	0.45	-0.03	-0.16	1
education_level	-0.03	0.23	-0.13	-0.18	0.01	1	-0.16
sex_male	0.04	-0.03	0.07	0.09	1	0.01	-0.03
age	0.01	0.14	-0.06	1	0.09	-0.18	0.45
avg_repay_status	0.28	-0.32	1	-0.06	0.07	-0.13	-0.03
limit_bal	-0.15	1	-0.32	0.14	-0.03	0.23	0.1
default_num	1	-0.15	0.28	0.01	0.04	-0.03	0.03

RQ Finalization: all 3 paths can determine the outcome

1 Demographics Path

2 Volume Path

3 Dynamic Path

To what extent can a borrower's default be predicted using demographics, credit limit, and past repayment behavior, and which of these three paths contributes most to predictive performance?

Empirical Study

Initial Model Specification and Early Results

1 Demographics Path

```
Call:
glm(formula = default_num ~ age + sex_male + education_level +
    married, family = binomial, data = df_corr)

Coefficients:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)     -1.106581   0.075219 -14.711  < 2e-16 ***
age             -0.002451   0.001713  -1.431    0.152
sex_male         0.205329   0.028459   7.215 5.40e-13 ***
education_level -0.101036   0.018943  -5.334 9.62e-08 ***
married          0.145496   0.031460   4.625 3.75e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 31673  on 29964  degrees of freedom
Residual deviance: 31569  on 29960  degrees of freedom
AIC: 31579

Number of Fisher Scoring iterations: 4
```

Initial Model Specification and Early Results

1 Demographics Path

+

2 Volume Path

```
Call:
glm(formula = default_num ~ age + sex_male + education_level +
    married + limit_bal, family = binomial, data = df_corr)

Coefficients:
                    Estimate   Std. Error z value             Pr(>|z|)
(Intercept)     -1.169635852  0.076398781  -15.31 < 0.0000000000000002 ***
age              0.004512040  0.001713174    2.63               0.0084 **
sex_male         0.168767346  0.028881387    5.84       0.000000005113 ***
education_level  0.054217785  0.020386058    2.66               0.0078 **
married          0.216989816  0.032023876    6.78       0.000000000012 ***
limit_bal       -0.000003512  0.000000133  -26.41 < 0.0000000000000002 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 31673  on 29964  degrees of freedom
Residual deviance: 30779  on 29959  degrees of freedom
AIC: 30791

Number of Fisher Scoring iterations: 4
```

Initial Model Specification and Early Results

1 Demographics Path

+

2 Volume Path

+

3 Dynamic Path

```
Call:
glm(formula = default_num ~ age + sex_male + education_level +
    married + limit_bal + avg_repay_status, family = binomial,
    data = df_corr)

Coefficients:
                     Estimate   Std. Error z value             Pr(>|z|)
(Intercept)      -1.546962189  0.080078328  -19.32 < 0.0000000000000002 ***
age               0.006550439  0.001770321    3.70              0.00022 ***
sex_male          0.097623468  0.029969376    3.26              0.00112 **
education_level   0.108474226  0.021476045    5.05        0.00000043964 ***
married           0.210231593  0.033110509    6.35        0.00000000022 ***
limit_bal        -0.000001964  0.000000136  -14.46 < 0.0000000000000002 ***
avg_repay_status  0.654127377  0.016656042   39.27 < 0.0000000000000002 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 31673  on 29964  degrees of freedom
Residual deviance: 29022  on 29958  degrees of freedom
AIC: 29036

Number of Fisher Scoring iterations: 4
```

Utrecht University

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shaping tomorrow